

San Jose State University

Design of a Wind Farm

ME 172: Alternative and Renewable Energy Sources

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In this project, an offshore wind farm in Cape Cod is being considered, as there is a location in the Nantucket Sound where shallow waters would allow for easier and cheaper construction. The proposed project will consist of 130 Siemens 3.6 MW wind turbines, which are commonplace for offshore wind projects. Should it be implemented, it would supply about 4% of the projected 2014 electricity demands for Massachusetts. In this study, the site's wind resources will be investigated using a yearlong dataset recorded close to the site. In addition to this, the electricity generated by the site will be calculated, as well as its expected capacity factor. Lastly, an economic analysis will be performed on the wind farm to determine the internal rate of return on the project.

Actual wind measurements performed by the Wind Energy Center in 2010 were used to characterize the wind resource of the Nantucket Sound location. The equipment used was mounted on the Bishop & Clerks lighthouse 15 meters above sea level. Available data consists of wind speed and direction recorded every 10 minutes from January 1st, 2010 at 12:00 AM until December 31, 2010 at 11:50 PM. The appropriate wind shear coefficient was used to extrapolate the data at 15 meters to the reference height of 50 m, and again to the wind turbine height of 80 m. The average wind speed for the year at 15, 50, and 80 meters was found to be 7.95, 8.98, and 9.38 meters respectively. According to the table on slide 43 of the "Wind Resources 2022" slidedeck provided by Francois Rongere, the Nantucket Sound site could be classified as a wind power class of seven.

The actual distribution of wind speeds at 80 m was graphed with wind speed bins of .5 m/s as seen in Figure 1.

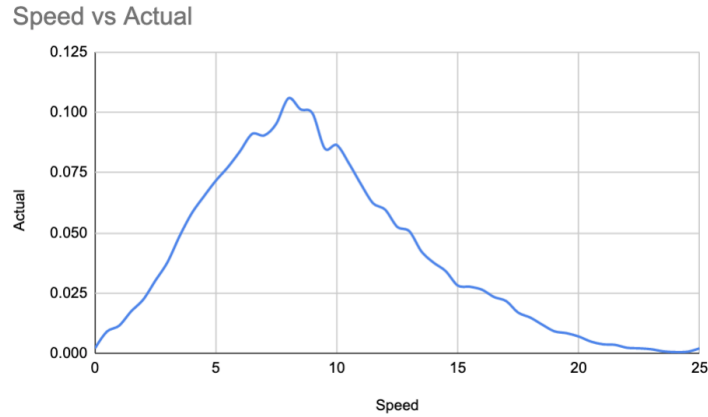


Figure 1: Actual wind speed distribution of Nantucket Sound for the year 2010 at intervals of .5 m/s.

Next, the Rayleigh's distribution function was calculated based on the average wind speed for the year at 80 m, again with wind speed bins of .5 m/s as seen in Figure 2.

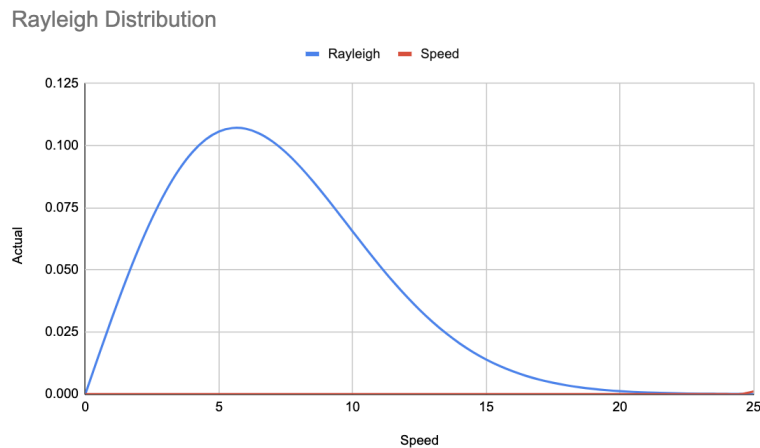


Figure 2: Rayleigh distribution based on the yearly average wind speed at 80 m for Nantucket Sound.

Comparing this Rayleigh distribution to the actual wind speed distribution, it can be stated that the Rayleigh distribution is fairly accurate, but seems to peak at a lower wind speed than the actual distribution. However, this observation allows for the conclusion that the site in reality is performing better than the Rayleigh distribution. The Weibull

distribution was also calculated and graphed alongside both the Rayleigh distribution and the actual wind speeds, as shown in Figure 3.

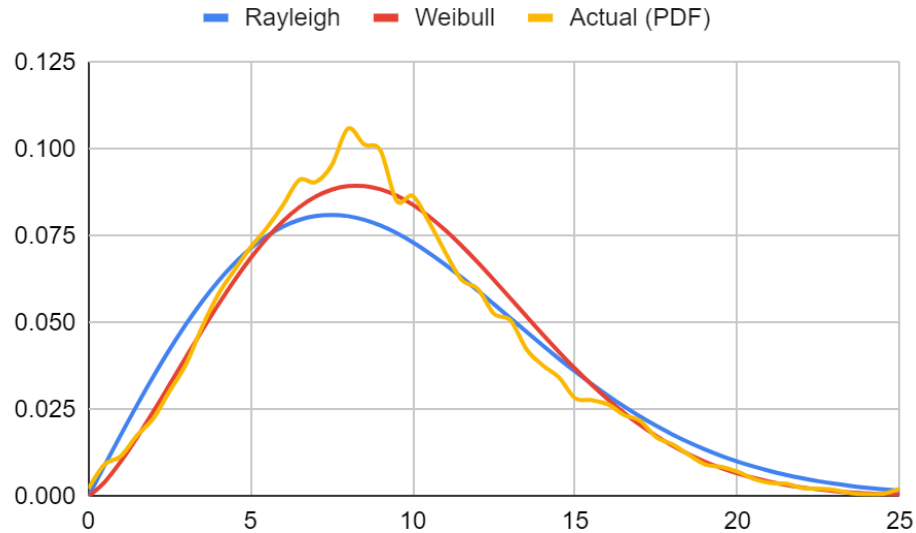


Figure 3: Weibull, Rayleigh, and actual distributions of wind speeds for Nantucket Sound at a height of 80 m.

Based on the shape and peak of the Weibull distribution, it can be considered a better fit than the Rayleigh distribution, as its curve is much more fitting to that of the actual distribution of wind speeds. In addition to this, the actual distribution of the wind azimuth in increments of 10° was calculated, assuming that wind direction does not change with altitude. This distribution can be seen in Figure 4.

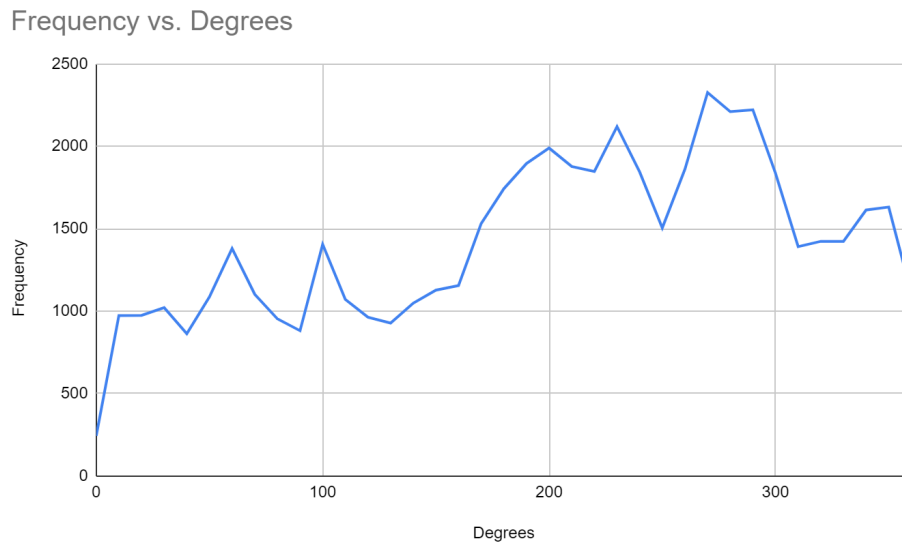


Figure 4: Actual distribution of wind azimuth for Nantucket Sound with wind direction bins of 10°.

Based on this distribution, it can be concluded that the Bishop & Clerks lighthouse measurement is accurate due to most of the wind coming from the south west and west.

To perform electricity generation calculations, it was provided that the 3.6 MW Siemens wind turbines would have hub height of 80 m and a diameter of 107 m. The power curve was provided, which shows electricity generation as a function of wind speed at 80 m, shown in Figure 5.

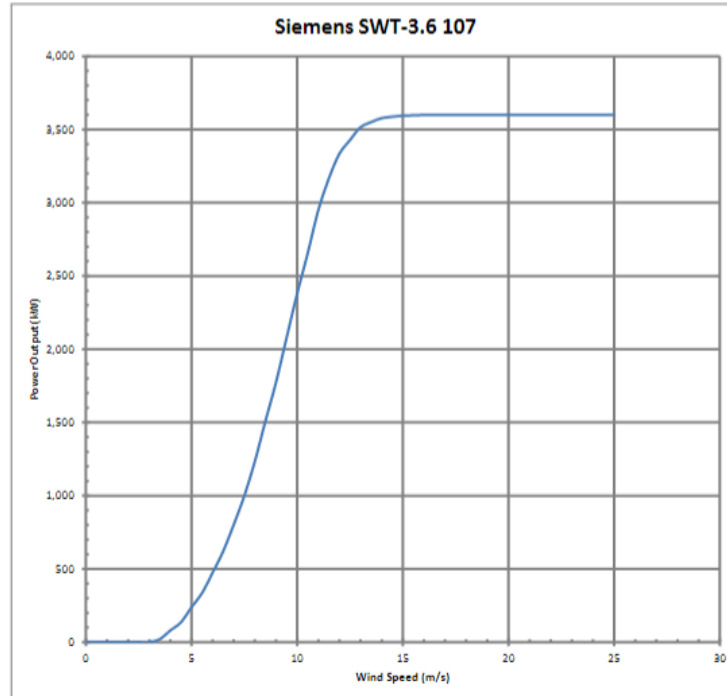


Figure 5: Power curve for Seimens SWT-3.6 107 wind turbine.

The power conversion rate of the turbine was plotted as a function of wind speed. This allowed for the determination of what wind speed resulted in the highest power generation. Based on the chart generated, it can be seen that the wind turbine produced the most electricity at wind speeds of about 8 m/s. The previously calculated actual wind distribution was used to find the distribution of power generated by the turbines, shown in Figure 6. Using these calculations, the annual energy delivered by one turbine was found to be 15.48 GWh, with a capacity factor of .49. With 130 turbines, the total energy delivered by the entire wind farm would be 2013.3 GWh.

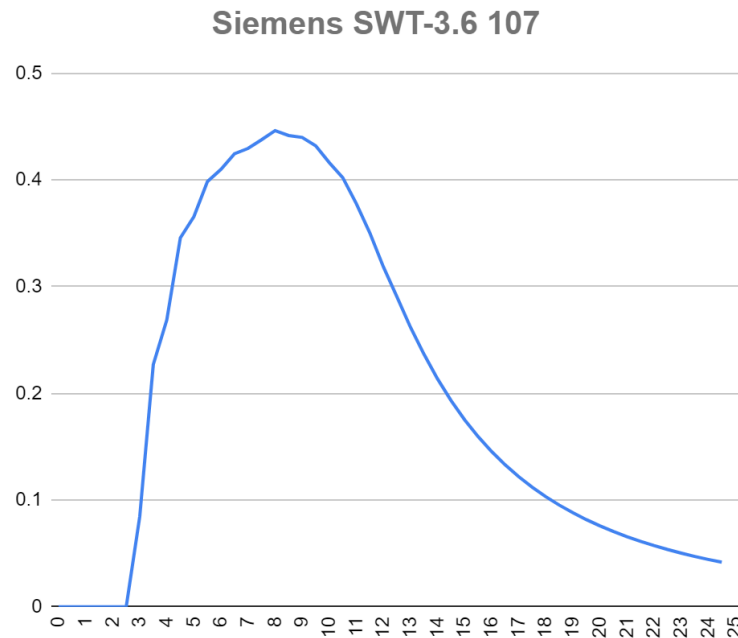


Figure 6: Power generation with wind speed bins of .5 from 0 to 25 m/s.

To perform the financial analysis on the wind farm, it was anticipated that the full cost of the farm would be \$4M/MW. Using New England's market price of electricity of \$.061/kWh, the internal rate of return over 25 years was calculated to be 4.23%. The internal rate of return was also calculated assuming that the project developer signed a PPA at \$0.187/kWh with local utilities, National Grid, and NSTAR for 77% of the farm's generated electricity. It was also assumed that the remaining 23% was sold at New England's market price. In this scenario, the internal rate of return on the wind farm rose to 17% over 25 years.

For the overall impact of the project on New England's electric system, it was important to note that the demand and consumption for New England were 26.5 GW and 133 TWh/y respectively. Using the annual 2013.3 GWh electricity generation of the wind farm calculated earlier, it can be determined that the wind farm would account for 1.5% of New England's annual electricity demand.